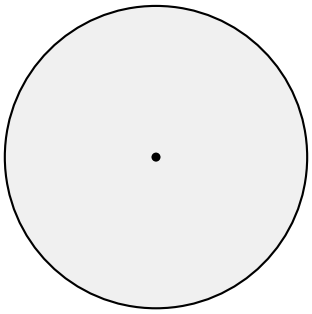
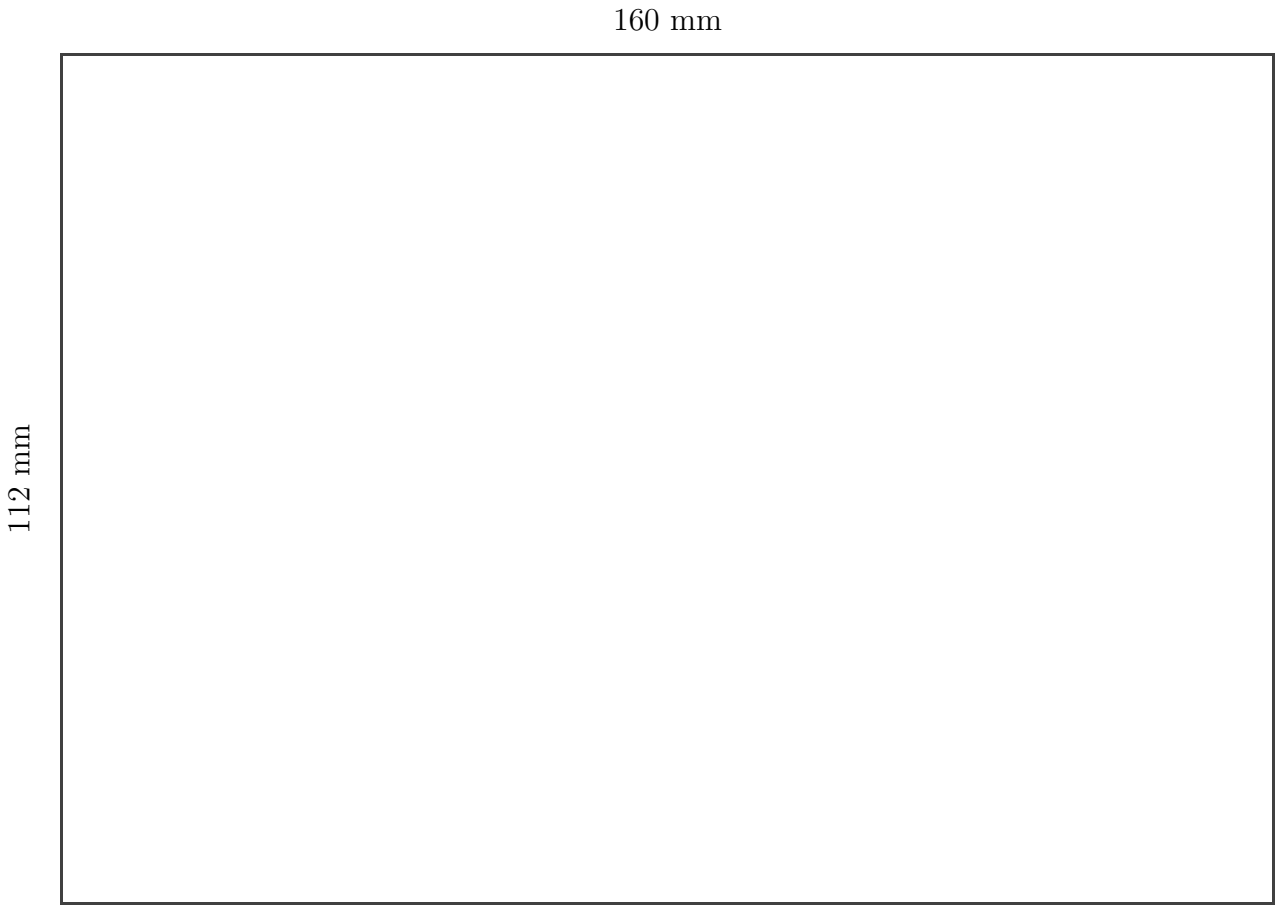
Disk sizes measure straight across the center. Keep disks flat. They may touch, but must not overlap. In a ring, every outside disk must touch the middle disk.

Problem 1: Use the 40 mm disks. How many can touch one middle disk at once? Make the biggest ring you can.



Most outside disks in our ring: _____

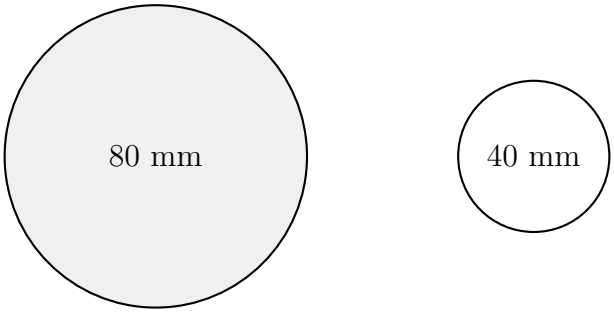
Problem 2: How many 40 mm disks can you fit inside this tray? Make a packing that holds as many as you can.



Practical tray: 160 mm × 112 mm interior

Our best packing holds _____ disks.

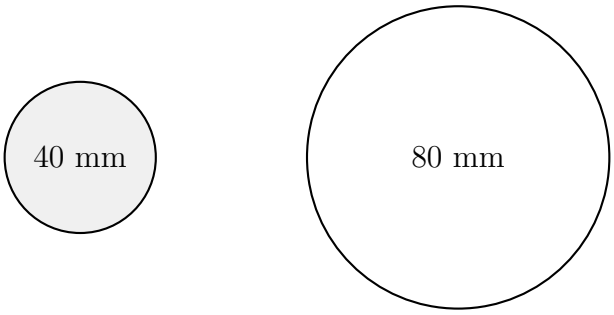
Problem 3: Put an 80 mm disk in the middle of your desk mat. Make the biggest ring of 40 mm disks that can touch it.



Sketches, not actual size

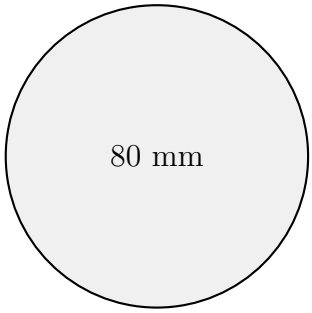
Most outside disks: _____

Problem 4: Now put a 40 mm disk in the middle. Make the biggest ring of 80 mm disks that can touch it.



Most outside disks: _____

Problem 5: Make the biggest ring you can using both 40 mm and 80 mm disks on the outside. Try each middle disk below. Every ring must have at least one outside disk of each size.



Middle disk

Most outside disks:

_____



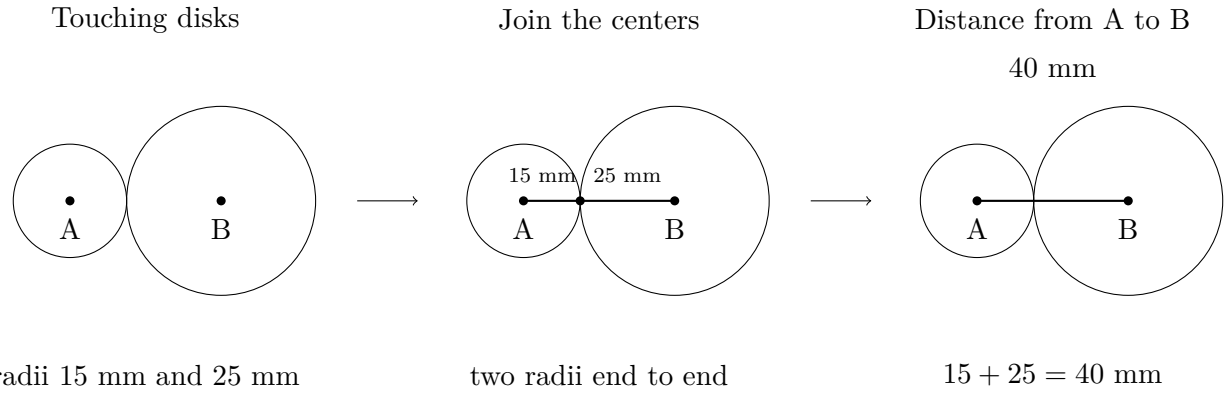
Middle disk

Most outside disks:

_____

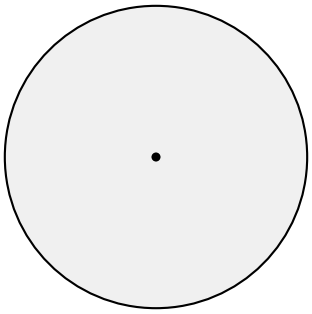
Sketches, not actual size. Build on your desk mat.

A radius runs from a circle's center to its edge. For two touching disks, the distance between centers is the sum of their radii.

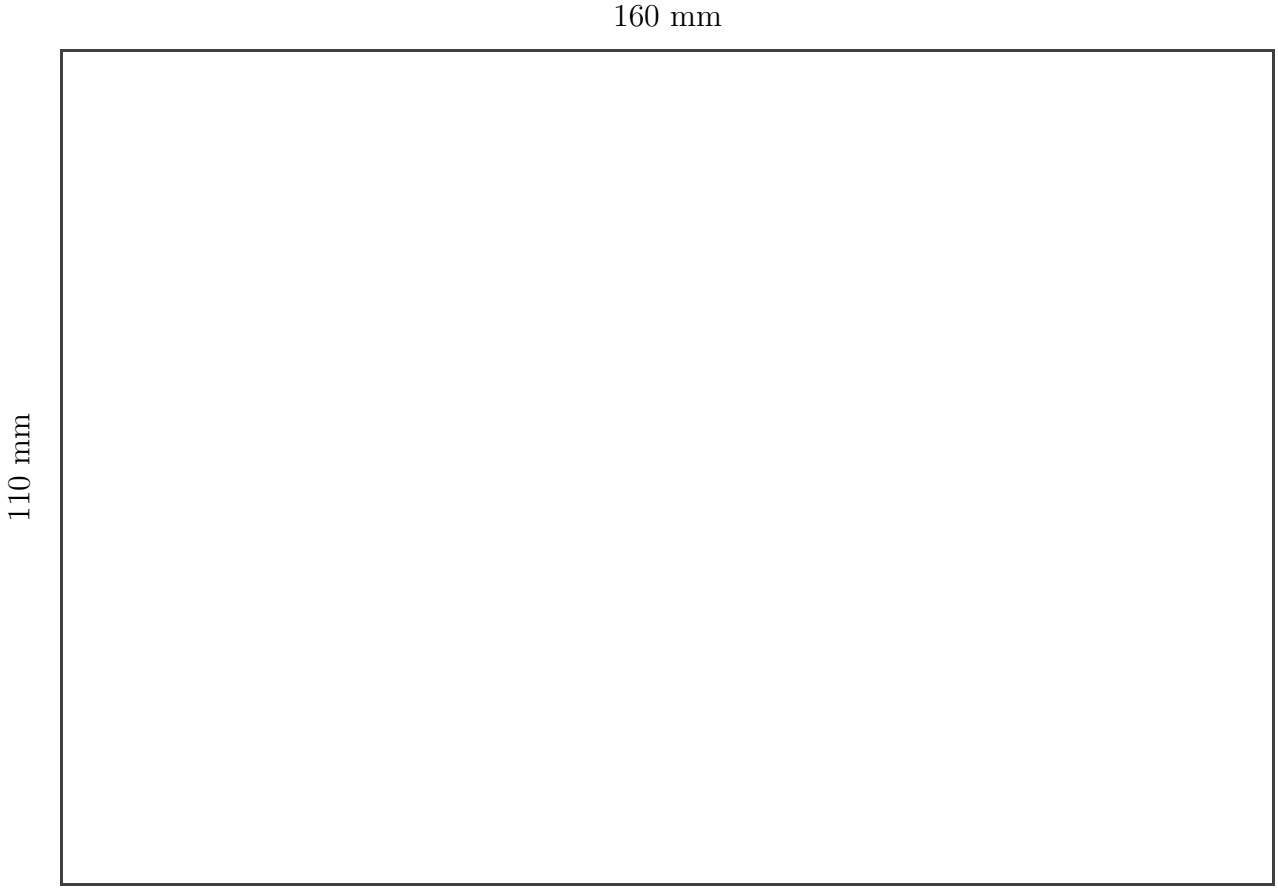


Same pair each time. Sketches, not actual size.

Problem 6: What is the largest possible ring of 40 mm disks around a 40 mm disk? Explain why one more cannot fit. In every largest ring, must neighboring outside disks touch each other?

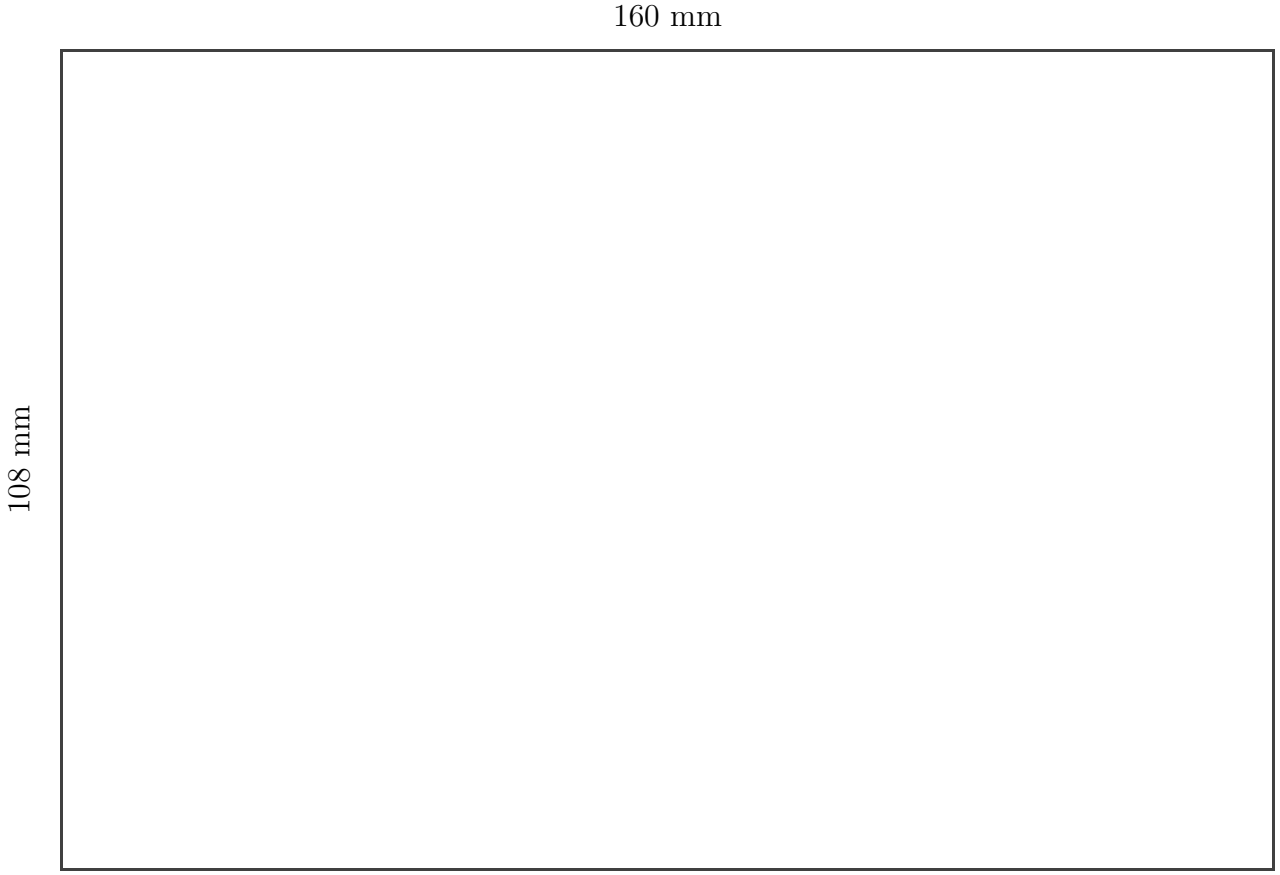


Problem 7: Fit eleven 40 mm disks in this exact tray, in three straight horizontal rows: 4, then 3, then 4. Use the center distances to decide whether the disks really fit.



Exact tray: 160 mm × 110 mm interior

Problem 8: Can eleven 40 mm disks fit in this tray in three straight horizontal rows: 4, then 3, then 4? Explain your answer for this row arrangement.

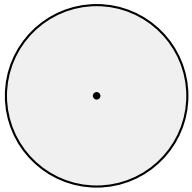


Exact tray: 160 mm × 108 mm interior

Problem 9: Keep the tray 160 mm wide. What is the smallest height that fits those three rows? Give an exact height if you can.

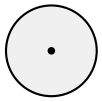
Problem 10: Return to your rings in Problems 3–5. For each one, can you explain why a ring with more outside disks is impossible? Draw the center distances you need.

40 mm outside



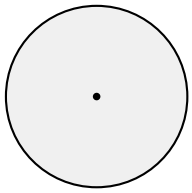
80 mm middle

80 mm outside



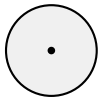
40 mm middle

Both sizes outside



80 mm middle

Both sizes outside



40 mm middle

Sketches, not actual size

Problem 11: When all the outside disks have the same size, what rule links the two disk sizes to the largest possible ring?